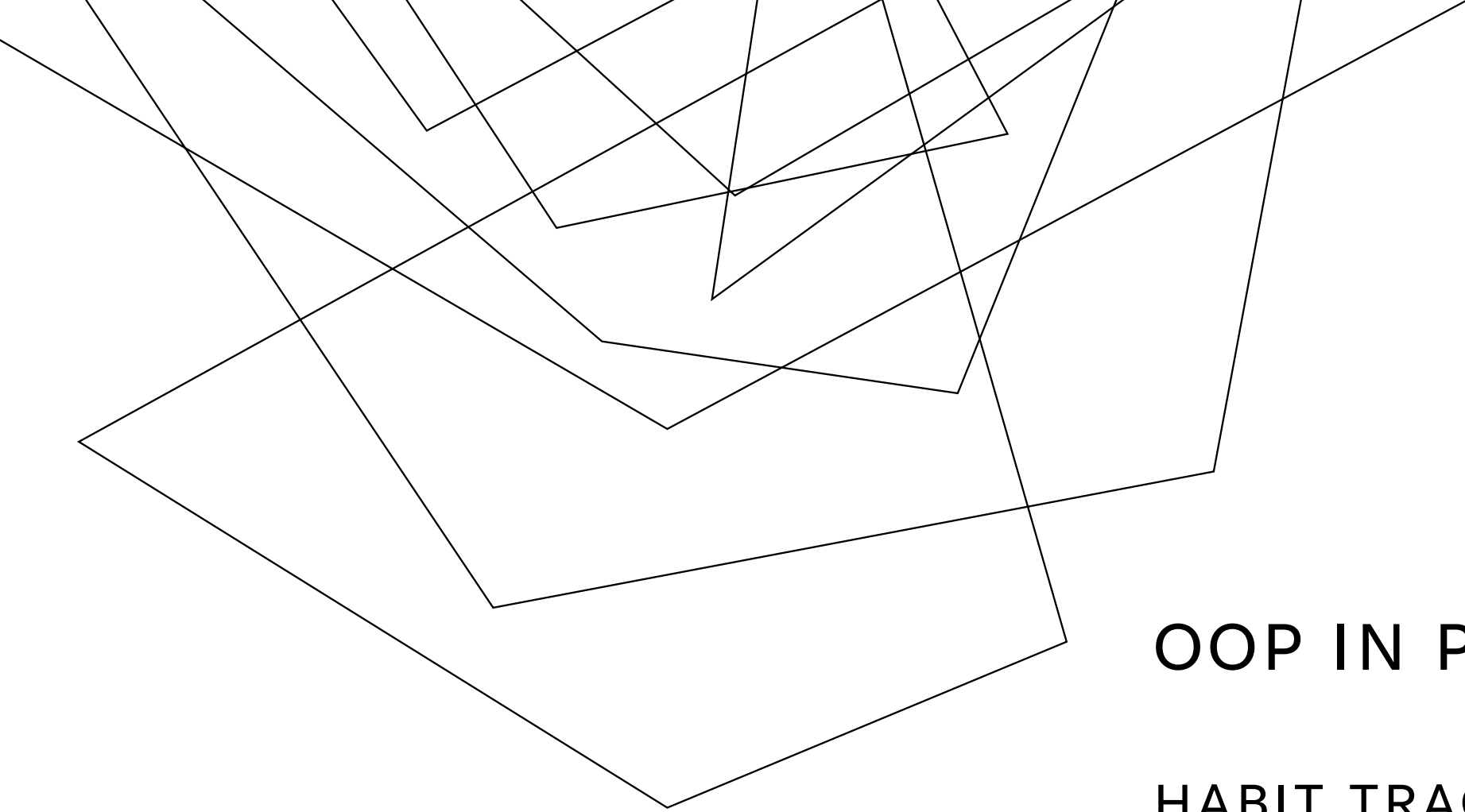


Abstract geometric lines in the top-left corner of the slide, consisting of several overlapping, irregular polygons and lines in black, creating a complex, layered effect.

OOP IN PYTHON

HABIT TRACKER

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AGENDA

What is the Habit Tracker?

Implementation

Operation

Results

Long-term consistency trumps short-term intensity
– Bruce Lee

WHAT IS THE HABIT TRACKER?

- The difficulty with staying consistent is keeping good habits
- The habit tracker allows:
 - Creation of habits (Daily or Weekly)
 - Tracking of habits
 - Completion of habits on a regular interval
- The application allows for this through
 - Simple, non-distracting interfaces
 - Powerful and robust coding techniques

Name	Period	Number	Start Date	Complete	Last Completed	Streak
+++++						
test_day	daily	1	2025-03-14 12:58:02	0	2025-03-14 12:58:02	1
test_week	weekly	2	2025-03-08 12:58:02	0	2025-03-10 12:58:02	1
test_daily_streak	daily	3	2025-03-08 12:58:02	0	2025-03-14 12:58:02	7
test_weekly_streak	weekly	4	2025-02-15 12:58:02	0	2025-03-14 12:58:02	4
test_extra	daily	5	2025-03-14 12:58:02	0	2025-03-14 12:58:02	1

IMPLEMENTATION



Code Structures

- Python 3.11 used
- Unit tests supplied with test database
 - Implemented through pytest
- Database instead of text files
 - Implemented with SQLite3
- Command Line Interface(CLI)
 - Implemented with Click
 - User-friendly menu and help options

Code Techniques

- OOP implemented
 - Ensuring efficient operation
- Modular programming implemented, ensuring:
 - Easy reading
 - Easy understanding



IMPLEMENTATION

Procedure

- Create a Class of habits with relevant information
 - Name, start date and time, period etc
 - May need library functions like datetime
- Create a database .py file for storing the information about a habit using SQLite3
- Create methods to solve each of the aspects required
- Create the CLI .py file using Click in order to expose an interface
- Start unit testing using a test database and data to run the tests using Pytest

OPERATION

Operations of the tracker

1. Main Menu
 - a. Help menu
2. Add Habits
3. Delete Habit
4. Tick Habit (Complete)
5. Display Habits (All)
6. Display Habits by period (Daily or weekly)
7. Display the highest streak of all habits
8. Display highest streak within a period

```
Input Habit Command > help add-habit
Usage: add-habit [OPTIONS]

Adds a habit to the database. The user is required to enter a habit name and
period :return: N/A: Calls a function that adds the habit, period and
default values to the DB

Options:
  --help  Show this message and exit.
Input Habit Command >
```

1. Main Menu

```
Welcome to your Habit terminal!

Here you can see to all your habitual needs. Type "home" for further instructions.

To exit the application: type "exit"

Input Habit Command > |
```

1a. Help Menu

```
Documented commands (type help <topic>):
=====
add-habit      display-habit      display-high-streak-period  home
delete-habit   display-high-streak  display-period-habit       tick-habit

Undocumented commands:
=====
exit  help  quit

Input Habit Command >
```

1a. Help Menu for “add-habit”

OPERATION CONTINUED

Each user input is validated and verified. 2 Add habit

```
Input Habit Command > add-habit
Please input the name of your new habit:  tester
You entered tester
Is this correct? Y/N:  y
Please input the periodicity of this habit:  daily
Habit successfully created
Input Habit Command >
```

3. Remove habit

```
Please input the name of the habit to be removed: tester
You entered "tester"
Is this correct? Y/N: y
Please input the periodicity of the habit to be removed: daily
Habit successfully removed
```

4. Tick Habit

```
Input Habit Command > tick-habit
Please input the name of the habit you completed: tester
You entered "tester"
Is this correct? Y/N: y
Please input the periodicity of the habit you completed: daily
Habit successfully completed. Well done!
```

OPERATION CONTINUED

Display operations

5. Display all habits

Name	Period	Number	Start Date	Complete	Last Completed	Streak
+++++						
test_day	daily	1	2025-03-14 12:58:02	0	2025-03-14 12:58:02	1
test_week	weekly	2	2025-03-08 12:58:02	0	2025-03-10 12:58:02	1
test_daily_streak	daily	3	2025-03-08 12:58:02	0	2025-03-14 12:58:02	7
test_weekly_streak	weekly	4	2025-02-15 12:58:02	0	2025-03-14 12:58:02	4
test_extra	daily	5	2025-03-14 12:58:02	0	2025-03-14 12:58:02	1

5. Display all habits by period

Name	Period	Number	Start Date	Complete	Last Completed	Streak
+++++						
test_day	daily	1	2025-03-14 14:13:18	0	2025-03-14 14:13:18	1
test_daily_streak	daily	3	2025-03-08 14:13:18	0	2025-03-14 14:13:18	7
test_extra	daily	5	2025-03-14 14:13:18	0	2025-03-14 14:13:18	1

6. Highest streak of all habits

The habit with the total longest streak is: test_daily_streak with 7 completions
Input Habit Command >

7. Highest streak by period

The daily with the longest streak is: test_daily_streak with 7 completions
Input Habit Command >

RESULTS

Unit Tests run

- Test_display_db → Displays a test Database
- Test_create_habit → Creates a test habit
- Test_remove_habit → Removes the test habit
- Test_complete_habit → Completes a habit in the database
- Test_recompleted_habit → Attempts to recomplete a habit (fails to update)
- Test_display_periodicity → Displays habits by period
- Test_longest_streak → Displays the habit with the longest streak
- Test_longest_streak_periodicity → Displays habit with the longest streak according to period

```
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-8.3.5, pluggy-1.5.0
rootdir: C:\Users\Saintsburg\PycharmProjects\Habit_Tracker1
collected 8 items

Test.py ..... [100%]

===== 8 passed in 0.15s =====
(venv) PS C:\Users\Saintsburg\PycharmProjects\Habit_Tracker1>
```

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract pattern on the left side of the slide.

THANK YOU

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